

5.3.25 SPI characteristics

Unless otherwise specified, the parameters given in Table 67 are derived from tests performed under the ambient temperature, f_{PCLKx} frequency and supply voltage conditions summarized in Table 18.

- Output speed set to $\text{OSPEEDRy}[1:0] = 11$
- Capacitive load $C_L = 30 \text{ pF}$
- Measurement points done at $0.5 \times V_{\text{DD}}$ level
- I/O compensation cell activated

Refer to Table 48. I/O static characteristics for more details on the input/output alternate function characteristics (NSS, SCK, MOSI, MISO for SPI).

Table 67. SPI characteristics

Evaluated by characterization - Not tested in production.

Symbol	Parameter	Conditions		Min	Typ	Max	Unit
f _{SCK}	SPI clock frequency	Master mode		-	-	72	MHz
		Slave receiver mode				100	
		Slave mode transmitter/full duplex				32	
t _{su} (NSS)	NSS setup time	Slave mode		3	-	-	ns
t _h (NSS)	NSS hold time			1	-	-	
t _w (SCKH), t _w (SCKL)	SCK high and low time	Master mode		T _{PLCK} - 1	T _{PLCK}	T _{PLCK} + 1	
t _{su} (MI)	Data input setup time	Master mode	DDRS = 0	2.5			
			DDRS = 1	10-T _{SCK} /2	-	-	
t _{su} (SI)		Slave mode		2	-	-	
t _h (MI)	Data input hold time	Master mode	DDRS = 0	1	-	-	
			DDRS = 1	(T _{SCK} /2)-7	-	-	
t _h (SI)		Slave mode		1.5	-	-	
t _a (SO)	Data output access time	Slave mode		12	13.5	15.5	
t _{dis} (SO)	Data output disable time	Slave mode		6.5	8.5	10.5	
t _v (SO)	Data output valid time	Slave mode		-	12	15.5	
		Master mode		-	2	2.5	
t _h (SO)	Data output hold time	Slave mode,		8	-	-	
t _h (MO)		Master mode		0	-	-	